📢 Aero Club Training Program Session Schedule

📅 15/06/2026 (Monday)

🕖 7:00 PM – 8:30 PM
Introduction

Topics Covered:
Introduction to Aero Club
Basics of Drone Design

Show more

Here's a cleaner and more professional version of the announcement:

Message

📢 **Aero Club Training Program – Session Schedule**

📅 **Date:** 15 June 2026 (Monday)

🕖 **7:00 PM – 8:30 PM**
Introduction Session

Topics Covered:

- Introduction to Aero Club
- Basics of Drone Design

🕣 **8:30 PM – 9:00 PM**
Aerodynamics Session

Topic Covered:

- Fundamentals of Aerodynamics

📌 The meeting link will be shared shortly before the session.

We encourage all interested juniors to attend and explore the exciting fields of drone and aircraft design, aerodynamics, avionics, and ongoing Aero Club projects.

For any queries, feel free to ask in the group or contact us personally.

This version is suitable for WhatsApp, Discord, Telegram, or an official club announcement.

see the topics covered in this by them so you also starting teaching me this what they have mentioned from depth i should get crystal clear concept and each term u explain pls

Absolutely. Let's treat this like the first lecture of an aerospace engineering and drone design course. I'll explain every concept from first principles, assuming no prior knowledge, and build up layer by layer.

Roadmap of What We'll Learn

From your Aero Club schedule:

1. Introduction to Aero Club → What aerospace engineering actually is
2. Basics of Drone Design
3. Fundamentals of Aerodynamics

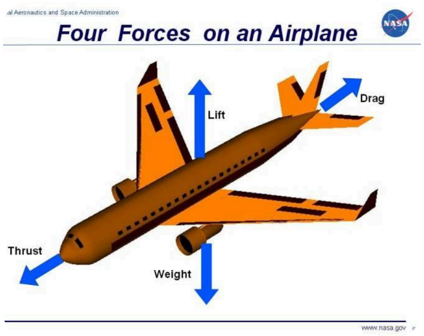
Before discussing drones or airplanes, you need to understand one fundamental thing:

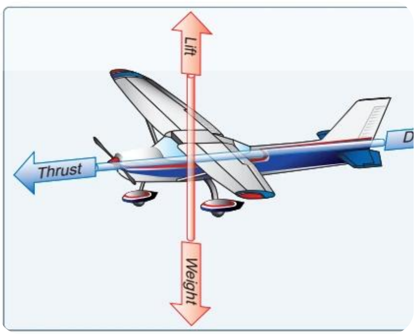
What is Flight?

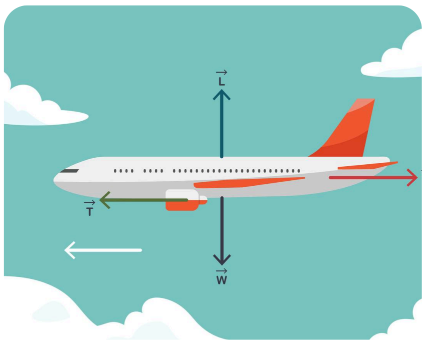
Flight is the act of overcoming gravity using forces generated by moving through air.

Everything in aerospace revolves around understanding and controlling forces.

When an aircraft flies, four forces act on it:







1. Weight (Gravity)

Weight is the force pulling the aircraft downward.

Formula:

Weight = Mass × Gravity

Example:

If a drone weighs 2 kg:

Where:

- kg = mass
- N (Newton) = unit of force

Gravity is always trying to pull the drone to Earth.

2. Lift

Lift is the upward force generated by airflow.

Lift is what fights gravity.

If:

Lift > Weight → aircraft climbs

Lift = Weight → aircraft maintains altitude

Lift < Weight → aircraft descends

A drone's propellers create lift directly.

An airplane's wings create lift.

3. Thrust

Thrust is the force pushing the aircraft forward.

Generated by:

- Propellers
- Jet engines
- Rocket engines

Without thrust:

No airflow over wings

No lift

No flight

4. Drag

Drag is air resistance.

It opposes motion.

Whenever something moves through air:

Air pushes back.

Examples:

- Stick your hand out of a moving car.
- Feel the force pushing your hand backward.

That force is drag.

First Important Aerospace Principle

Every aircraft designer is trying to solve:

Generate maximum lift with minimum drag.

This single sentence summarizes most of aerodynamics.

What Is Aerodynamics?

Let's break the word.

Aero = Air

Dynamics = Motion/Forces

Aerodynamics = Study of how moving air interacts with objects.

Examples:

- Aircraft

- Drones
- Cars
- Birds
- Rockets

Whenever air moves around something, aerodynamics is involved.

What Is Air Actually?

Many beginners think air is "nothing."

Wrong.

Air is matter.

It has:

- Mass
- Weight
- Pressure
- Density

Air is mainly:

- 78% Nitrogen
- 21% Oxygen
- Other gases

Since air has mass, it can exert forces.

That is why airplanes can fly.

The Concept of Pressure

This is one of the most important concepts in aerospace.

Pressure means:

Force acting on an area.

Formula:

Pressure = Force / Area

Unit:

Pascal (Pa)

Imagine:

You place a book on a table.

The book presses down on the table.

That pressing effect is pressure.

Air also exerts pressure.

Atmospheric Pressure

The atmosphere is a huge column of air above us.

That air has weight.

Therefore it creates pressure.

At sea level:

Atmospheric pressure ≈ 101,325 Pa

or

101.3 kPa

You don't feel crushed because the pressure inside your body balances the outside pressure.

What Happens When Air Moves Faster?

This leads us to one of the most famous concepts in aerodynamics.

Bernoulli's Principle

It states:

As the speed of a fluid increases, its pressure decreases.

Think of it like:

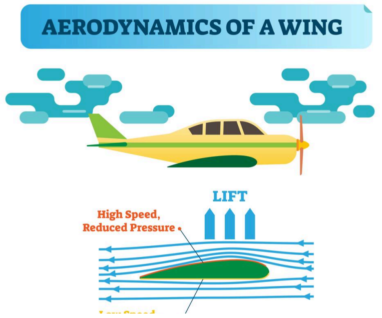
Slow air → High pressure

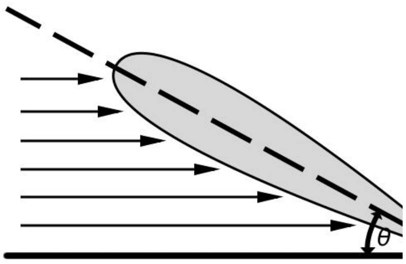
Fast air → Low pressure

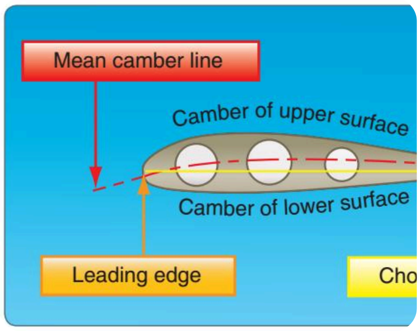
This principle helps explain wing lift.

How Wings Generate Lift

A wing shape is called an airfoil.







Notice:

- Top surface is curved
- Bottom surface is flatter

As air flows around the wing:

Air on top moves faster.

Air below moves slower.

Therefore:

Top pressure becomes lower.

Bottom pressure remains higher.

Result:

Higher pressure below pushes upward.

That upward force is lift.

But There Is An Even Better Explanation

Modern aerodynamics also uses Newton's laws.

The wing pushes air downward.

Air pushes the wing upward.

This is:

Newton's Third Law

"For every action there is an equal and opposite reaction."

Wing pushes air down.

Air pushes wing up.

Lift is created.

This explanation is extremely important for drone propellers too.

How Drone Propellers Work

A propeller is basically a rotating wing.

Most students miss this.

A propeller blade is actually an airfoil.

When spinning:

- Blade pushes air downward.
- Air pushes blade upward.

This creates thrust/lift.

For a quadcopter:

All four propellers together generate enough lift to overcome weight.

If lift exceeds weight:

Drone rises.

What Is a Drone?

A drone is an unmanned aircraft.

Official term:

UAV (Unmanned Aerial Vehicle)

Components:

1. Frame
2. Motors
3. Propellers
4. Flight Controller
5. Battery
6. ESCs (Electronic Speed Controllers)
7. Sensors

Think of it as:

Body + Muscles + Brain + Nervous System

Drone Component Breakdown

Frame

The skeleton.

Holds everything together.

Needs:

- Strong
- Light

Common materials:

- Carbon fiber
- Aluminum
- Plastic

Motors

The muscles.

Convert electrical energy into rotational motion.

Measured using:

KV Rating

Example:

1000 KV motor

Means:

1000 RPM per volt

at no load.

If supplied 10 V:

≈ 10,000 RPM

Propellers

The wings.

Convert motor rotation into thrust.

Larger propeller:

- More thrust
- Better efficiency

Smaller propeller:

- Faster response
- Less thrust

Battery

Energy source.

Usually:

Lithium Polymer (LiPo)

Example:

3S Battery

3 cells in series

Each cell:

3.7 V nominal

Therefore:

3S = 11.1 V

ESC (Electronic Speed Controller)

Acts as translator.

Flight controller says:

"Motor 1 → 50% speed"

ESC converts that command into electrical power for the motor.

Flight Controller

The brain.

Examples:

- Pixhawk
- Betaflight boards
- Ardupilot systems

Reads sensors and decides motor speeds.

Why Doesn't A Drone Flip Over?

Because of sensors.

Mainly:

Gyroscope

Measures rotation.

Tells drone:

"You're tilting left."

Accelerometer

Measures acceleration.

Tells drone:

"You're moving."

Flight controller uses these sensors hundreds of times per second.

It continuously adjusts motor speeds.

That is why modern drones appear stable.

The Most Important Concept For Beginners

Everything in aerospace boils down to controlling:

1. Forces
2. Motion
3. Energy

If you understand:

- Lift
- Weight
- Thrust
- Drag
- Pressure
- Airflow
- Newton's laws

then 80% of beginner aerodynamics becomes intuitive.

Quick Check

Can you answer these three questions?

1. Why does a drone rise when motor speed increases?
2. Why does an airplane need forward motion to keep flying?
3. If drag increases but thrust stays the same, what happens to the aircraft?

Try answering them yourself. Once you do, I'll start **Lecture 2: Drone Design in Depth**, where we'll cover:

- Why quadcopters have 4 motors
- Motor KV and propeller selection
- Frame geometry
- Center of gravity
- Stability
- Flight controller logic

- How professional drones are actually designed from scratch.